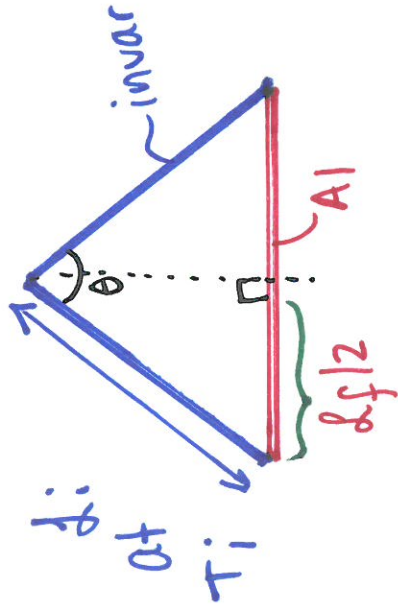


Problem 19-54



$$\begin{aligned}
 T_i &= 0^\circ\text{C} \\
 l_i &= l_0 \quad (\text{for both Al and invar}) \\
 \Delta T &= T_f - T_i \\
 &= T_c - 0^\circ\text{C} \\
 \Delta T &= T_c
 \end{aligned}$$

(a) To start $\theta = 60^\circ$. (equilateral triangle)

From the construction:

$$\begin{aligned}
 \sin\left(\frac{\theta}{2}\right) &= \frac{l_{f,Al}/2}{l_0} \quad \leftarrow \text{Al length changes} \\
 &= \frac{l_0(1 + \alpha_{Al}\Delta T)/2}{l_0} \quad \leftarrow \text{invar length is fixed} \\
 &= \frac{l_0(1 + \alpha_{Al}T_c)}{2l_0} \\
 \theta &= 2 \sin^{-1}\left(\frac{1 + \alpha_{Al}T_c}{2}\right)
 \end{aligned}$$

(c) If $T_c = 0^\circ\text{C}$

$$\Rightarrow \theta = 2 \sin^{-1}\left(\frac{1}{2}\right)$$

$$\theta = 2(30^\circ) = 60^\circ \quad \text{What you expect.}$$

(b) If $T_c < 0^\circ\text{C}$, the angle θ will be less than 60° .

(d) Now both lengths change:

$$\sin\left(\frac{\theta}{2}\right) = \frac{\cancel{L_0}(1 + \alpha_{Al}\Delta T)/2}{\cancel{L_0}(1 + \alpha_{Invar}\Delta T)}$$

$$\Rightarrow \sin\left(\frac{\theta}{2}\right) = \frac{1 + \alpha_{Al}T_c}{2(1 + \alpha_{Invar}T_c)}$$

(e) Al melts at 660°C

Invar melts at 1427°C

Use $T_c = 660^\circ\text{C}$

$$\begin{aligned}\theta &= 2 \sin^{-1} \left[\frac{1 + \alpha_A T_C}{2(1 + \alpha_{\text{invar}} T_C)} \right] \quad \text{--- ①} \\ &= 2 \sin^{-1} \left[\frac{1 + (24 \times 10^{-6} / ^\circ\text{C})(660^\circ\text{C})}{2[1 + (0.9 \times 10^{-6} / ^\circ\text{C})(660^\circ\text{C})]} \right]\end{aligned}$$

$$\theta = 61^\circ$$

(f) For $T_C = -273^\circ\text{C}$ use eq. ① above:
 $\theta = 59.6^\circ$